

18 C

$$x = \frac{7.5 \times 10^{-3}}{9.5}$$

$$\frac{\lambda}{d} = \frac{x}{D} \Rightarrow \lambda = d \left(\frac{x}{D} \right) = 0.45 \times 10^{-3} \left(\frac{7.5 \times 10^{-3}}{0.70 \times 9.5} \right) = 5.08 \times 10^{-7} \text{ m}$$

19 D $d \sin \theta = m \lambda$, where $\sin \theta = 1$ at max. order